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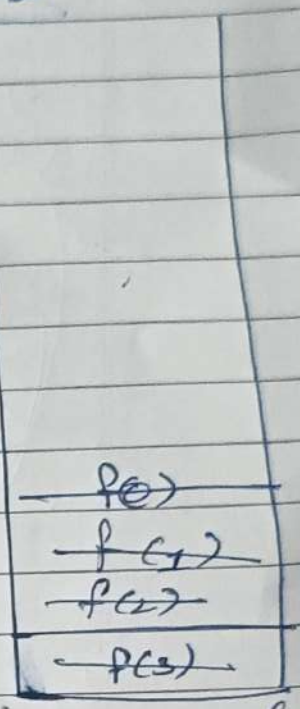
```
Void fun (int n) {
    if (n==0) return
    Syso(n+" "); → 3
    fun (n-1)
    Syso (n+1 +" "); → 4
}

Void fun (int n) {
    if (n==0) return;
    Syso (n +"  "); → 2
    fun (n-1);
    Syso (n+1 +" ");

Void fun (int n) {
    if (n==0) return;
    Syso (n+" ") → 1
    fun (n-1)
    Syso (n+1 +" ").

void fun (int n) {
    if (n==0) return;
```

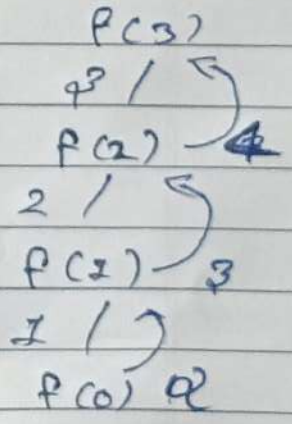


call stack

output

3
2
1
2
3
4

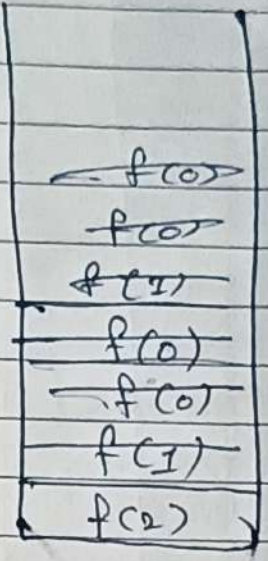
recurssive tree



(2)

```
Void f (int n) {
  if (n==0) return
  Syso (X);
  f(n-1)
  Syso (Y);
  f (n-1)

Void f (int n) {
  if (n==0) return
  Syso (X);
  f(n-1);
  Syso (Y);
  f (n-1);

Void f (int n) {
  if (n==0) return
  Syso (X);
  f (n-1);
  Syso Y
  f (n-1)
repeat f (0)
Void f (int n) {
  if (n==0) return
  Syso (X);
  f(n-1)
  Syso (Y);
  f (n-1)

Void f (int n) {
  return

Void f (int n) {
```

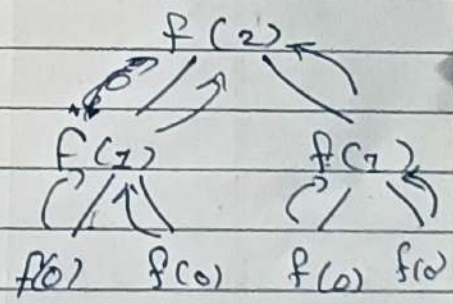
Call stack



Output

X
X
Y
Y
X

recursive tree



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(3)

```
Void f (int n) {   (4)
    if (n <= 1) {
        Syso (n)
    return ;
}
fun (n-2) ; -> 4-2 = (2)
    Syso (n)
fun (n-2) ;
}

Void f (int n) {   (2)
    if (n <= 1) {
        Syso(n)
        return ;
}
fun (n-2); -> 2-2 = 0
    Syso (n) -> print (2)
fun (n-2)

Void f (int n) {   (0)
    if (n <= 1) {
        Syso (n) -> 0
        return
}
void f (int n) {   (0)
    if (n <= 1) {
        Syso (n) -> 0
        return
}
void f (int n)   (2)
```

~~f(0)~~
~~f(0)~~
~~f(2)~~
~~f(0)~~
~~f(0)~~
~~f(2)~~
~~f(4)~~

output
0
2
0
4
0
2
0

(4)

```
Void f (int n) {
    if (n==0){
        return
    }
    fun(n-1);
    Syso(n*n);

    fun(n-1)

    Void f (int n) {
        if (n==0){
            return
        }
        fun(n-1);
        Syso(n*n+" ");

        fun(n-1)

        Void f (int n) {
            if (n==0){
                return
            }

            Void f (int n) {
                if (n==0){
                    return
                }

                Void f (int n) {
                    if (n==0){
                        return
                    }
```

Call stack	Output
~~f(0)~~	1
~~f(0)~~	4
~~f(1)~~	.
~~f(0)~~	1
~~f(0)~~	
~~f(1)~~	
~~f(2)~~	

```
fun(n-1)
    Syso(n*n)
    fun(n-1)

    Void f (int n) {
        if(int n==0){
            return
        }
        fun (int n) {
            (int n==0){
```

(5)

```
Void f (int n) {
    if (n ≤ 0) {
        return;
    }
    Syso (n + " ");  → 5
    f(n-2)
    Syso(n + " ");  → 5

Void f (int n) {
    if (n ≤ 0) {
        return;
    }
    Syso (n + " ");  → 3
    f(n-2)
    Syso (n + " ");  → 3

Void f (int n) {
    if (n ≤ 0) {
        return;
    }
    Syso (n + " ")  → 1
    f(n-2)
    Syso (n + " ")  → 1

Void f (int n) {
    if (n ≤ 0) {
        return
    }
```

call stack
~~f(1)~~
f(3)
f(5)

output
5
3
1
1
3
5

Recurrsive tree.

f(5)
/ ↑
f(3)
/ ↑
f(1)

6.)

```
Void fun (int n) {                    (8)
    if (n <= 1) {
        Syso (n+" ");
        return;
    }
Syso (n+" ");  → 8
    fun (n/2);
Syso (n+" ");

Void fun (int n) {                    (4)
    if (n <= 1) {
        Syso (n+" ");
        return;
    }
Syso (n+" ")  → 4
    f(n/2);
Syso (n+" ")

Void fun (int n) {                    (2)
    if (n <= 1) {
        Syso(n+" ");
        return
    }
Syso(n+" ");  → 2
    f(n/2);
Syso(n+" ");

Void fun (int n) {                    (0)
    if (n <= 1) {
        Syso (n+" ")
        return
    }
```

Call stack

~~f(0)~~
~~f(2)~~
~~f(4)~~
~~f(8)~~

Output

8
4
2
0
2
4
8

f(8)
|
f(4)
|
f(2)
|
f(0)

7.) Void fun (int n) {

```
if (n <= 0) return;
  Syso (A + n + " ");
    f (n-1);
  Syso (B+(n*2) + " ");
}
Void fun (int n) {
    if (n <= 0) return;
     Syso (A + n + " ");
      f (n-1);
     Syso (B+ (n*2)+" ");

Void fun (int n) {
    if (n <= 0) return;
     Syso (A + n + " ");
     f (n-1);
     Syso (B+ (n*2)+" ");

Void fun (int n) {
    if (n <= 0) return
```

Call stack

f(0)
f(1)
f(2)
f(3)

output

A3
A2
A1
B2
B4
B6

recurssive tree

f(3)
/
f(2)
/
f(1)
/
f(0)

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(8) void fun (int n) {

```
    if (n <= 0) return;
      fun (n-1);
    Syso (n + " ");
      fun (n-2);
  }

fun (int n) {
    if (n <= 0) return;
      fun (n-1);
    Syso (n + " ");
      fun (n-2);

fun (int n) {
    if (n <= 0) return;
      fun (n-1);
    Syso (n + " ");
    fun (n-2);

fun (int n) {
    if (n <= 0) return;
      fun (n-1);
    Syso (n + " ");
      fun (n-2);

fun (int n) {
    if (n <= 0) return;

fun (int n) {
    if (n <= 0) return;

fun (int n) {
    if (n <= 0) return;
}
```

Page (2)

~~f(0)~~
~~f(-1)~~
~~f(0)~~
~~f(1)~~
~~f(2)~~
~~f(-1)~~
~~f(0)~~
~~f(1)~~
~~f(0)~~
~~f(-1)~~
~~f(0)~~
~~f(1)~~
~~f(2)~~
~~f(0)~~
~~f(0)~~

output:
1
2
3
1
1
(2)

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Page

```
fun (int (n) 1 ){
    if (n<=0) return
    fun (n-1)
    Syso(n+" ");
    fun (n-2)

fun (int (n) 0 ){
    if (n<=0) return

fun (int n -1 ) {
    if (n<=0) return

* fun (int n 2 ){ *
    if (n<=0) return
    fun (n-1);
    Syso (n+" ");
    fun (n-2); *

fun (int n 1 ){
    if (n<=0) return
    fun (n-1);
    Syso (n+" ");
    fun (n-2);

fun (int n 0 ){
    if (n<=0) return
    ~~fun(n-1)~~;
    ~~Syso (n+" ");~~
    ~~fun~~

fun (int n -1 ){

    if (n<=0) return;
```

```
fun (int n 0 ){
    if (n<=0) return
```

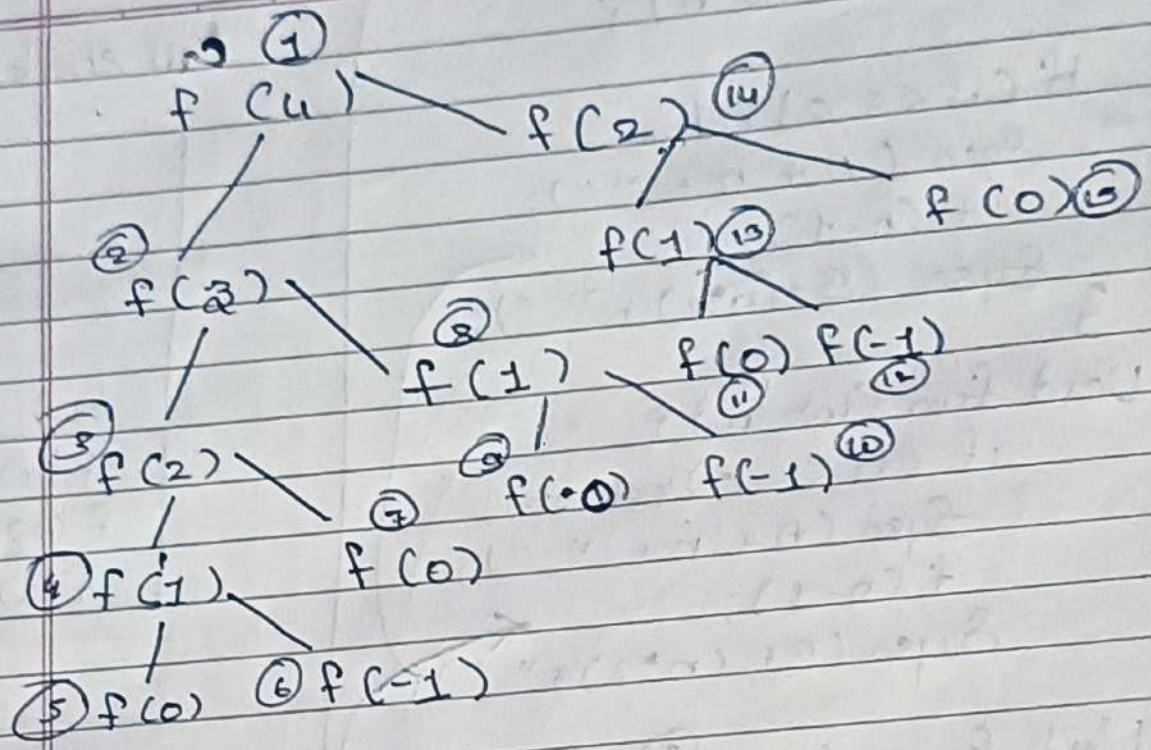
9

```
Void fun (int n)3 {
 if (n <= 1) {
    Syso (n + " ");
    return
 }
 fun (n-1);
 Syso (n)
 fun (n-2);

Void fun (int n)2 {
 if (n <= 1) {
    Syso (n + " ");
      return;
 }
 fun (n-1);
 Syso (n)
 fun (n-2);

Void fun (int n)1 {
 if (n <= 1) {
    Syso (n) ---> 1
    return
 }

Void fun (int n)0 {
 if (n <= 1) {
    Syso (n) -> 0
    return
 }

Void fun (int n)1 {
 if (n <= 1) {
    Syso (n) ---> 1
    return
 }
```

~~f(0)~~
~~f(1)~~
~~f(2)~~
f(3)

output
1
2
0
3
1

10.



Total 15 function calls.